

James Munday

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Academic Positions

University of Potsdam, Post-doctoral Research Fellow

Feb 2026 – present

University of Warwick, Research Fellow

June 2025 – Jan 2026

Education

University of Warwick, Doctor of Philosophy in Physics. Award: 30/09/2025

Oct 2021 – Sept 2025

University of Surrey, MPhys in Physics with Astronomy, First Class Honors

Sept 2017 – June 2021

Instituto de Astrofísica de Canarias (IAC), Master's degree research placement

Feb 2020 – Dec 2020

Teaching and leadership roles

Support Astronomer Isaac Newton Group - La Palma, Spain

Nov 2022 – Nov 2023

- I operated the 2.5 m Isaac Newton Telescope while working with the Isaac Newton Group.
- I taught the telescope to professors, post-doctoral or doctoral researchers, and post-graduate visitors. I advised how to optimise observations for their scientific goals.
- I performed telescope maintenance and instrumental setup changes.

Undergraduate laboratory demonstrator, University of Warwick – Coventry, UK

Oct 2021 – May 2022

- I led an experiment for 2nd year physics and astronomy undergraduates, and I graded their laboratory books and reports after the experiment.
- The experiment involved using spectrographs on small telescopes to measure the radial velocity of the Sun using absorption lines to learn the usage and calibration of spectroscopic data.

Publications

H-index: 16. Number of citations from refereed papers: 977, starting in Oct. 2020.

50 refereed publications (3 more in press), of which I was first author in eight and second author in two. 1 first author white paper. Three first author papers feature press releases. My ADSABS library is linked with this text. Journal referee for A&A and MNRAS.

First author publications:

- **V407 Vul: a triple star system with an AM CVn detectable by gravitational wave observatories.** Submitted to A&A
- **Classifying white dwarfs from multi-object spectroscopy surveys with machine learning.** In MNRAS (Mar. 2026). arXiv: 2602.04964
- **The DBL Survey II: towards a mass-period distribution of double white dwarf binaries.** In MNRAS (Aug. 2025). arXiv: 2507.14123
- **A super-Chandrasekhar mass type Ia supernova progenitor at 49 pc set to detonate in 23 Gyr.** In Nature Astronomy (Apr. 2025). arXiv: 2504.04522
- **The DBL Survey I: discovery of 34 double-lined double white dwarf binaries.** In MNRAS (Aug. 2024), arXiv: 2407.02594
- **An Eclipsing 47 minute Double White Dwarf Binary at 400 pc.** In MNRAS (Aug. 2023), arXiv: 2308.00036
- **Two decades of optical timing of the shortest-period binary star system HM Cancr.** In MNRAS (Feb. 2023), arXiv: 2211.09834
- **The post-common-envelope binary central star of the planetary nebula ETHOS 1.** In MNRAS (Sept. 2020),

arXiv: 2009.03577

First author white paper:

- **ESO Expanding Horizons White Paper: Electromagnetic characterisation of millihertz gravitational-wave sources in the Milky Way.** Submitted to the ESO Expanding Horizons Call for White Papers. arXiv: 2507.14123

Supervised and co-supervised students

Master's student: Shreyas Kathuria, University of Potsdam. <i>White dwarf binaries</i>	April 2026 – Present
Master's student: Ruchika Rani, University of Potsdam. <i>Machine learning techniques with multi-object spectroscopy surveys</i>	June 2026 – Present

Awards

Highly Commended for the Michael Penston Thesis Prize (2026)

Invited seminars

- 2026: Invited talk at University of Hamburg's astronomy group's colloquium
- 2025: Invited talk in the Rocky Worlds: WD Planetary research group at the University of Cambridge
- 2025: Invited talk at the University of Potsdam
- 2025: Invited talk in the multiple stars research group at the University of Amsterdam

Conferences

- 4MOST All Hands Meeting 2026 – attendee
- Tautenburg-Bamberg-Potsdam meeting 2026 – speaker
- 10 years to LISA (JPL) – invited speaker
- Progress in Astrophysics with Type Ia Supernovae third meeting (PATIAS-3) – speaker and attendee
- LISA Astrophysics Working Group Meeting 2024 – speaker and attendee
- 23rd European Workshop on White Dwarfs (EUROWD24 Barcelona) – speaker and attendee.
- LSST UK workshop series - attendee
- University of Warwick machine learning workshop - attendee
- AM CVn 4.5 (virtual) – session chair, speaker and attendee.
- 22nd European Workshop on White Dwarfs (EUROWD22 Tubingen) – speaker and attendee.
- UK National Astronomy Meeting (NAM) 2022 – attendee.
- Birmingham LISA astronomy work group conference 2022 – speaker and attendee.
- Spanish Astronomical Society 2020 (virtual) – attendee and slides on ETHOS 1.

Awarded Telescope Time

Principle Investigator (PI) – Total telescope time (metre class): 2m) 28 nights. 4m) 6 nights. 8m) 11 nights.

- **INT/2.5 m, IDS:** 28 nights [9 + 10 + 10 staff time]. **SALT/9.2 m, RSS+HRS:** 56 hours + 21 hours
- **CAHA35/3.5 m, CARMENES:** 22 hours. **GTC/10.4 m, HiPERCAM:** 7 hours + 2 hours
- **VLT/8.2 m, FORS2+X-Shooter+UVES:** 23 hours + 4.5 hours + 4.5 hours. **WHT/4.2 m, WEAVE:** 40 hours
- **NTT/3.6 m & TNO/2.5 m,:** a few nights of guaranteed time observations on ULTRACAM and ULTRASPEC

Co-Investigator – Total telescope time (metre class): 1m) 26 nights 2m) 43 nights. 4m) 4 nights. 8m) 2 nights.

- **VLT/8.2 m, Xshooter:** 4.5 hours. **NTT/3.5 m, ULTRACAM:** 4 nights. **GTC/10.4 m, HiPERCAM:** 13 hours.
- **INT/2.5 m IDS:** 32 nights [10 + 5 + 10 + 7]. **WHT/4.2 m, WEAVE:** 70 hours. **LCO/1 m, NRES:** 26 nights.
- **NOT/2.54 m ALFOSC+FIES:** 10 nights and 2 fast-track proposals of length 4 hours each [all proposals written by me]. Multiple exposures in staff time.

Software Experience

Python, Fortran, STARLINK astronomy package, Image Reduction and Analysis Facility (IRAF), Visual Basic (VB.net), Javascript, HTML and CSS. Experience in developing large pipelines from scratch in Python (see github.com/JamesMunday98/). Plenty of experience in managing raw photometric and spectroscopic data.

Outreach

- **4MOST Public Outreach:** I am an active contributor towards 4MOST collaboration news posts, social media engagement and newsletters.
- **Observatory sessions.** I welcomed public groups, particularly kids, in events at Warwick's campus observatory. I led a session teaching the purpose of a telescope and the science that can be done with one.
- **Physics outreach ambassador at the University of Surrey.** I regularly engaged in physics and astronomy outreach to the public at stargazing nights. I taught workshops to school classes age 12-15. I demonstrated a variety of aspects of physics in campus-hosted fairs aimed towards all ages.
- **General.** I led the La Palma journal club meeting that discusses current papers and topics in astronomy with staff members and PhD/Master level students. I actively shared astronomy with the public at the Roque de los Muchachos observatory. I taught schools of undergraduates visiting the Isaac Newton Telescope.

Languages

English: Native; Spanish: C1; German: A1

Certificates

- Spanish diploma as a foreign language, organisation DELE, level B2: Pass
(Diploma de Español como lengua extranjera DELE nivel B2: Apto)
- Global Graduate Award German Stage 2: Pass

References

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